

Decision-Invariant Sim-to-Real Vision-and-Language Navigation with Pseudo-Panoramic Observations

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Abstract

Following natural language instructions to complete navigation tasks is a crucial capability for real-world embodied robots. In vision-and-language navigation (VLN), agents typically assume access to complete, on-the-fly environmental observations and rely on them for decision making. However, most sim-to-real VLN approaches approximate the privileged complete panoramic sensing in simulation with monocular sensors, leading to significant semantic loss and incomplete perception. In this work, we present DIP², a sim-to-real VLN framework that reduces the discrepancy between assumed and realizable observations by introducing a unified sensing and mapping representation shared across simulated and real-world domains. Specifically, dense pseudo-panoramic observations are synthesized from a self-assembled surround-view camera system to recover rich semantics comparable to simulation, while a structure-only metric map unifies simulated RGB-D inputs with real-world LiDAR observations. Furthermore, to promote sim-to-real decision invariance, DIP² employs a learning-free local navigable waypoint prediction strategy via radial expansion, applied consistently in both domains. The predicted waypoints and dense semantic observations are integrated into a global topological map, allowing existing agents trained in simulation to be directly deployed in the real world. Extensive experiments in simulated and real environments demonstrate that DIP² substantially improves sim-to-real navigation performance. Source code will be published at <https://github.com/zheng19845/DIP2>.

CCS Concepts

• Computing methodologies → Robotic planning.

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Keywords

Vision-and-Language Navigation, Sim-to-Real Transfer, Multimodal Sensor Fusion, Cross-Domain Semantic Retrieval.

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1 Introduction

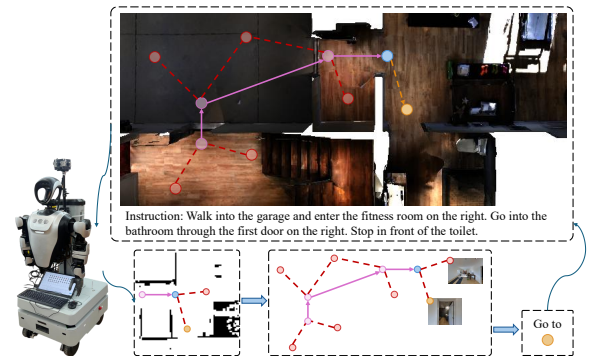


Figure 1: DIP² removes perceptual privilege by approximating panoramic observations with a practical sensor setup, enabling effective sim-to-real deployment of VLN agents.

Extensive research has been conducted in the field of vision-and-language navigation (VLN) [4] to enable real-world robots to interpret natural language and perform navigation tasks. A central goal of VLN is to develop embodied agents that can be deployed on physical robotic platforms, thereby supporting sim-to-real transfer. However, the majority of existing methods are developed and evaluated exclusively in simulation [1, 2, 4, 6, 8, 9, 11, 13, 14, 16–18, 20, 21, 25–27], either in a discrete (VLN) [4] or a continuous setting (VLN-CE) [15], leading to a pronounced sim-to-real gap.

As a matter of fact, sim-to-real VLN agents often experience substantial performance degradation compared to their simulated counterparts. A fundamental cause lies in the observation model assumed by VLN agents, which typically presumes access to complete, on-the-fly environmental observations and rely heavily on such information for decision making. However, most prior sim-to-real VLN approaches fail to provide such holistic information, as they adopt monocular sensing [7, 10, 12, 22–24] to approximate these privileged observations. While convenient, monocular inputs severely restrict the agent’s field of view and semantic coverage, leading to incomplete environmental understanding and increased perceptual ambiguity. Beyond perceptual limitations, effective sim-to-real transfer also critically depends on the invariance of the decision pipeline. Early work by Anderson et al. [3] sought to directly deploy simulated VLN agents by modifying or replacing key components at deployment time. Such changes introduce inconsistencies in the agent’s decision pipeline, further exacerbating the sim-to-real gap. These observations highlight the need to decouple perceptual inputs from action generation and to enforce a unified, domain-consistent intermediate representation.

In this work, we propose DIP², a **Decision-Invariant** sim-to-real VLN framework that addresses these challenges by synthesizing **Pseudo-Panoramic** observations from a realizable sensor setup and introducing a learning-free topo-mapping strategy to enforce consistency between simulation and the real world (as shown in Figure 1). To approximate holistic environmental information, DIP² leverages a self-assembled surround-view camera system to synthesize dense pseudo-panoramic viewpoints, recovering rich semantic context comparable to that available in simulation and enabling more reliable navigation decisions than monocular alternatives. In parallel, DIP² constructs a structure-only metric map that unifies simulated panoramic RGB-D inputs and real-world LiDAR observations into a shared, modality-agnostic representation, providing comprehensive structural information while reducing discrepancies induced by sensing differences. To further mitigate variance in the decision pipeline and promote sim-to-real behavioral consistency, we leverage this representation to introduce a learning-free local navigable waypoint prediction strategy based on radial expansion. By decoupling decision making from domain characteristics, this strategy enforces invariant decisions across sensing modalities while remaining agnostic to the observation source. Finally, the proposed local waypoints, together with the dense pseudo-viewpoints, are integrated into a global topological navigation map to allow for seamless transfer of simulated agents to real-world deployment.

Our main contributions are summarized as follows:

- We propose DIP², a sim-to-real VLN framework that addresses observation mismatch by approximating panoramic perceptual inputs using a low-cost surround-view camera system, enabling dense pseudo-panoramic semantic perception on real-world robotic platform.
- We introduce a modality-agnostic navigation representation that decouples perception from action by introducing a structure-only metric map with a unified waypoint generation strategy. This representation bridges simulated

panoramic RGB-D observations and real-world LiDAR inputs, enabling identical waypoint prediction and topological map construction across domains.

- We validate DIP² through extensive experiments in both simulated and real-world environments, demonstrating consistent improvements in navigation performance and a substantial reduction of the sim-to-real gap compared to existing sim-to-real VLN baselines.

2 Related Work

2.1 Vision-and-Language Navigation

Vision-and-Language Navigation aims to equip embodied agents with the ability to follow natural language instructions to navigate in previously unseen environments, a capability that is fundamental to the deployment of intelligent robots. Early VLN studies relied on photo-realistic simulators such as Matterport3D [5] to train agents to follow natural language instructions [4]. In this setting, navigation is defined on a predefined graph, where agents select directions and move between nodes. Early methods [4] used sequence-to-sequence models to map instructions to actions. Later, Fried et al. [11] introduced panoramic observations, enabling high-level decision-making without explicit low-level actions. Subsequent work further improved performance via imitation and reinforcement learning [20], topological memory [9], volumetric representations [17], and LLM-based reasoning [16, 25–27].

2.2 Vision-and-Language Navigation in Continuous Environments

More recently, many works focus on the VLN-CE task. Introduced by Krantz et al. [15], VLN-CE requires agents to follow natural language instructions in continuous indoor environments such as Habitat [19]. Unlike discrete VLN, it requires both high-level semantic reasoning and low-level continuous control. Most methods follow two approaches: (1) hierarchical frameworks that transfer discrete VLN designs via waypoint prediction from panoramic observations and low-level control [1, 2, 13, 14, 21], achieving strong performance; and (2) zero-shot methods using LLMs [6, 8, 18], which currently suffer from low success rates and high latency. However, both rely on idealized panoramic observations, which are difficult to obtain on real robots, making real-world deployment challenging and motivating further research.

2.3 Sim-to-Real Vision-and-Language Navigation

Compared to VLN and VLN-CE in simulation, relatively little work targets agents transferable to real-world environments. To approximate holistic observations, some studies restrict agents to monocular or limited field-of-view sensors [7, 10, 12, 22–24], but this leads to significant performance drops compared to panoramic-based methods. This is because panoramic perception provides richer semantic context and simplifies navigation as viewpoint selection. To address this, we propose DIP², which approximates comprehensive on-the-fly observations using a realizable sensor setup on a mobile robot. Unlike prior work that adapts policies to limited inputs,

our approach preserves the panoramic observation model during deployment, maintaining consistent semantics across domains.

3 Method

3.1 Problem Formulation & Overview

In this section, we briefly describe the task formulation of VLN-CE, as well as the pipeline of DIP².

In the VLN-CE task, each episode starts with the agent being positioned in a continuous 3D environment with an initial pose $\mathbf{s}_0 = (\mathbf{R}_0, \boldsymbol{\tau}_0)$, where $\boldsymbol{\tau}_0 \in \mathbb{R}^3$ denotes the agent's position and $\mathbf{R}_0 \in \text{SO}(3)$ denotes its rotation. The agent is additionally provided with a natural language navigation instruction $\mathbf{W} = \{\mathbf{w}_1, \dots, \mathbf{w}_H\}$. Conditioned on this instruction, the agent aims to reach the target location specified by $\mathbf{W}$ through a sequential decision-making process. At each time step t , the agent receives an egocentric visual observation $\mathbf{O}_t$ and selects a control action $a_t \in \mathcal{A}$, which induces a state transition of the agents position $\mathbf{s}_{t+1} = f(\mathbf{s}_t, a_t)$.

As discussed in Sec. 1, the primary challenge in deploying VLN agents from simulated environments to real-world settings arises from discrepancies in sensor observations. In simulation, the agent's observation at time step t can be formulated as

$$\text{sim}\mathbf{O}_t = \{\text{sim}\mathbf{V}_t, \text{sim}\mathbf{D}_t\}, \quad (1)$$

where $\text{sim}\mathbf{V}_t = \{\text{sim}\mathbf{v}_t^1, \dots, \text{sim}\mathbf{v}_t^K | \text{sim}\mathbf{v}_t^k \in \mathbb{R}^{W \times H \times 3}\}$ and $\text{sim}\mathbf{D}_t = \{\text{sim}\mathbf{d}_t^1, \dots, \text{sim}\mathbf{d}_t^K | \text{sim}\mathbf{d}_t^k \in \mathbb{R}^{W \times H \times 1}\}$ denote the RGB and the aligned depth observations at step t , respectively. These observations consist of $K = 12$ rendered views uniformly sampled around the agent.

In contrast, such dense multi-view RGB-D observations are typically unavailable in real-world sensors. Consequently, in this work, we investigate VLN deployment on a physical robot equipped with a LiDAR sensor and a surround-view system comprising four fisheye cameras. The real-world observation at time step t is formulated as,

$$\text{real}\mathbf{O}_t = \{\text{real}\mathbf{V}_t, \mathcal{P}_t\}, \quad (2)$$

where $\text{real}\mathbf{V}_t = \{\text{real}\mathbf{v}_t^1, \dots, \text{real}\mathbf{v}_t^L | \text{real}\mathbf{v}_t^l \in \mathbb{R}^{W' \times H' \times 3}\}$ consists of $L = 4$ surround-view images, and $\mathcal{P}_t \in \mathbb{R}^{N \times 3}$ denotes the point cloud collected from a LiDAR.

To enable effective sim-to-real transfer, DIP² approximates complete environmental information by synthesizing dense pseudo-panoramic views from a low-cost surround-view camera system, enabling rich semantics without specialized sensors. Simulated RGB-D and real-world LiDAR are unified into a modality-agnostic metric map for structural understanding. Based on this shared representation, DIP² performs learning-free waypoint prediction via radial expansion and integrates waypoints with semantic observations into a global topological map. This ensures a consistent decision pipeline, allowing agents trained in simulation to operate identically in real world. Figure 2 shows an overview of DIP².

3.2 Modality-Unified Preprocessing

3.2.1 Dense Pseudo Viewpoint Generation. To approximate a complete environmental observation as in simulations, we first construct a cylindrical panorama from a calibrated surround-view camera system and then re-project the panorama to synthesize intermediate viewpoints.

Specifically, all cameras are pre-calibrated with known intrinsics $\mathbf{K}_i$ and extrinsics $(\mathbf{R}_{\text{cam}_i}, \boldsymbol{\tau}_{\text{cam}_i})$ for the i -th camera defined with respect to a common world coordinate frame. Input images are first undistorted, and all projections are performed in the pinhole camera space.

Cylindrical Panorama Synthesis. Let the cylindrical panorama have resolution $W_p \times H_p$. For each panorama pixel, we associate a 3D point on a virtual cylinder of radius r . Specifically, for a panorama pixel $\mathbf{p} = [c_i, c_j]$, the corresponding 3D point $\mathbf{P}_w$ on the cylinder in the global frame is defined as,

$$\mathbf{P}_w = \begin{bmatrix} r \sin\left(2\pi \frac{c_i}{W_p}\right) \\ -\gamma \left(\frac{H_p}{2} - c_j\right) \\ r \cos\left(2\pi \frac{c_i}{W_p}\right) \end{bmatrix}, \quad (3)$$

where γ denotes the vertical scaling factor of the cylindrical surface. The world point $\mathbf{P}_w$ is then transformed into the camera coordinate frame and projected onto the image plane via,

$$\mathbf{p}_{\tilde{\text{cam}}_i} = \mathbf{K}_{\text{cam}_i} (\mathbf{R}_{\text{cam}_i} \mathbf{P}_w + \boldsymbol{\tau}_{\text{cam}_i}), \quad (4)$$

where $\mathbf{p}_{\tilde{\text{cam}}_i}$ denotes the homogeneous image coordinate for the i -th camera. The resulting pixel correspondence defines a dense mapping from the cylindrical panorama to each camera image, with which the panorama can be synthesized by sampling color values from the corresponding camera images.

Furthermore, to alleviate discontinuities in overlapping regions of the panorama caused by exposure and response differences among cameras, we apply linear alpha fusion in overlapping areas. Denote two camera images with an overlapping region after projection onto the cylindrical surface as $\mathcal{I}_a$ and $\mathcal{I}_b$, for each pixel $\mathbf{p}$ in the overlap region, the final panorama intensity is computed via linear alpha blending:

$$\mathcal{I}_p(\mathbf{p}) = \alpha(\mathbf{p}) \mathcal{I}_a(\mathbf{p}) + (1 - \alpha(\mathbf{p})) \mathcal{I}_b(\mathbf{p}), \quad (5)$$

where the weight $\alpha(\mathbf{p}) \in [0, 1]$ varies linearly along the horizontal axis, ensuring a smooth transition between adjacent views.

Pseudo Viewpoint Reprojection. After obtaining the cylindrical panorama, we synthesize K dense pseudo viewpoints in correspondence to simulation by uniformly sampling the azimuth angle. For the i -th pseudo view, a virtual pinhole camera is defined with a rotation $\mathbf{R}_y(\phi_i)$ along the y -axis and a 3D point $\mathbf{P}_c$ is projected onto the i -th pseudo view as,

$$\tilde{\mathbf{p}}^i = \mathbf{K}_v (\mathbf{R}_y(\phi_i) \mathbf{P}_c), \quad \phi_i = \frac{2\pi i}{K}, \quad (6)$$

where $\tilde{\mathbf{p}}^i$ denotes the homogeneous image coordinate in the i -th pseudo view and $\mathbf{K}_v$ is the intrinsic matrix of the virtual cameras.

3.2.2 Unified Egocentric Metric Mapping. To construct a unified representation for removing inductive bias, we adopt an egocentric metric map as a unified, structure-only representation for sim-to-real transfer. The resulting map encodes only geometric occupancy, without semantic labels, and can be constructed from both simulated RGB-D and real-world LiDAR inputs, ensuring cross-domain consistency.

Semantic-aided Metric Mapping in Simulated Environment. In simulation, due to the availability of panoramic RGB-D observations, semantic predictions from panoramic RGB-D observations

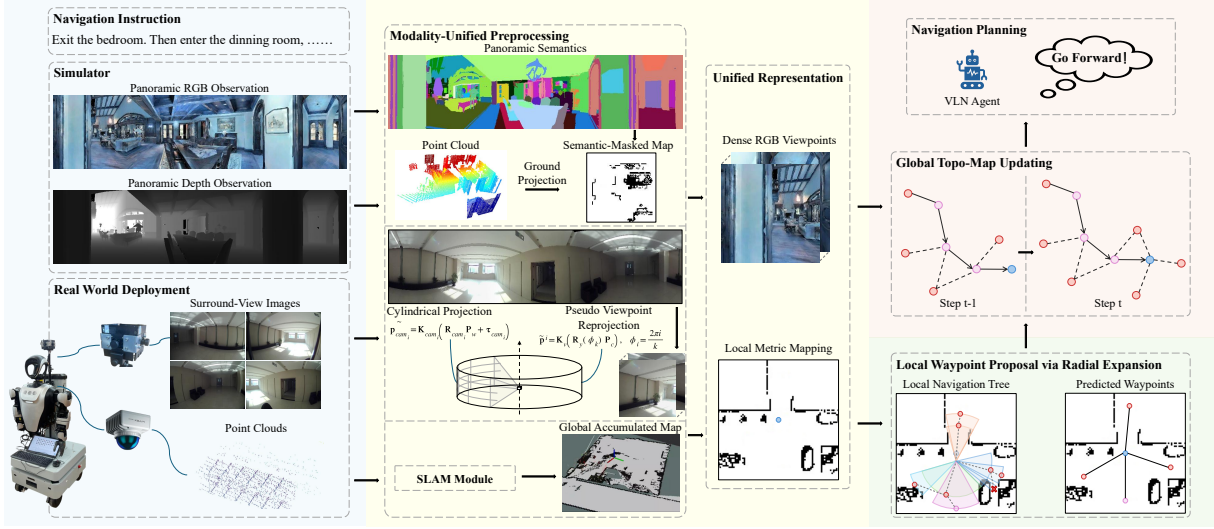


Figure 2: Overview of DIP². DIP² addresses sim-to-real VLN by emulating comprehensive on-the-fly observations and enforcing invariance in decision representations across domains. Heterogeneous sensor inputs are first transformed into a shared, modality-agnostic representation that combines dense semantics with a structure-only metric map. Next, a learning-free local waypoint prediction module generates navigable waypoints via radial expansion, inducing a consistent intermediate decision structure. Finally, the predicted waypoints and semantic observations are integrated into a unified global topological map, enabling VLN agents to be trained and deployed with identical decision pipelines in both simulation and real world.

are used to identify and mask out traversable regions during map construction. Specifically, we process the agent’s observation at each time step t . Given each RGB image $^{sim}v_t^i$ and its aligned depth map $^{sim}d_t^i$ from the i -th viewpoint at time step t , where the viewpoint is defined by a rotation R^i relative to the agent’s pose, we first apply semantic segmentation to obtain a dense per-pixel semantic prediction,

$$^{sim}s_t^i = f_{sem}\left(^{sim}v_t^i\right). \quad (7)$$

From $^{sim}s_t^i$, we construct a binary navigability mask $M_t^i \in \{0, 1\}^{H \times W}$. Utilizing the navigability mask, semantic categories corresponding to traversable regions are masked out,

$$M_t^i(u, v) = \begin{cases} 1, & ^{sim}s_t^i \notin C_{nav}, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

where C_{nav} denotes the set of navigable semantic classes (e.g., floor or stairs).

The navigability mask is then applied to the depth image to filter out invalid regions, yielding a masked depth map that contains depth measurements only for unnavigable obstacles,

$$^{sim}\tilde{d}_t^i(u, v) = M_t^i(u, v) \cdot ^{sim}d_t^i(u, v). \quad (9)$$

Then, for each valid pixel (u, v) with $^{sim}\tilde{d}_t^i(u, v) > 0$, we back-project the pixel into the agent’s current egocentric coordinate frame using the known camera intrinsics K_{sim} and rotation R^i :

$$P_t^{ego}(u, v) = R^i \begin{pmatrix} ^{sim}\tilde{d}_t^i(u, v) K_{sim}^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \end{pmatrix}, \quad (10)$$

where $P_t^{ego} \in \mathbb{R}^3$ denotes the 3D point in camera coordinates.

To obtain a 2D metric representation on the ground plane, we project P_t^{ego} onto the horizontal plane by discarding the vertical component. The resulting 2D points are discretized into a fixed-resolution grid and accumulated over the K cameras to form an egocentric metric map $M_t \in \mathbb{R}^{N \times N}$, where each cell encodes the occupancy of the corresponding spatial region around the agent. This semantic-aided mapping process explicitly integrates semantic priors with metric depth geometry, producing a noise-robust egocentric representation while only retaining structural information. **Mapping in Real-World Environment.** For real-world deployment, metric mapping is constructed using a SLAM system that incrementally estimates the agent pose $s_t = (R_t, \tau_t)$ and accumulates a global point cloud map $\mathcal{P}^{glob} = \bigcup_t \mathcal{P}_t$. At time step t , a local point cloud $\mathcal{P}_t^{loc}$ is extracted by transforming the global map into the agent-centric frame using the inverse pose transformation,

$$\mathcal{P}_t^{loc} = R_t^T (\mathcal{P}^{glob} - \tau_t). \quad (11)$$

The local point cloud is then projected onto the ground plane to obtain a 2D egocentric metric map $M_t \in \mathbb{R}^{N \times N}$, following the same procedure as in the simulated environment.

3.3 Adaptive Local Waypoint Prediction

We propose a unified waypoint prediction strategy across domains using the egocentric metric map to eliminate variance in the decision pipeline. Each waypoint is parameterized by a direction and step length relative to the agent’s pose. The proposed strategy is shown in Figure 3.

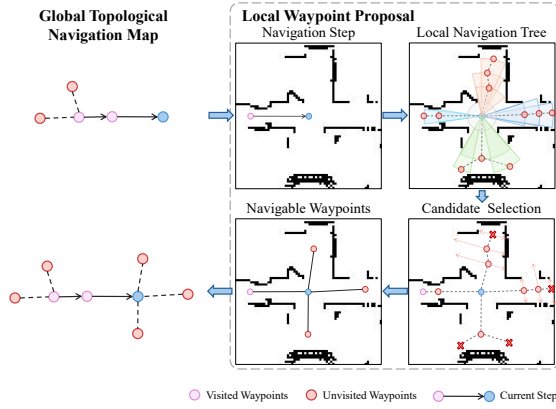


Figure 3: Illustration of the proposed adaptive local waypoint prediction. At each step, we construct a local navigation tree via a radial expansion strategy, followed by pruning to select valid waypoint candidates. The step length of each candidate is further refined to account for lateral exploration coverage.

3.3.1 Local Navigation Tree Construction via Radial Expansion.

First, we incrementally construct a local navigation tree using a radial search strategy with progressively increasing step lengths. Starting from a parent sector c , every radial search expansion partitions each sector into n contiguous angular sub-sectors $\{c'_1, \dots, c'_n\}$, where each sub-sector represents a candidate navigable direction within the parent sector. By recursively applying this radial partitioning process, the navigation tree captures locally feasible directions while progressively refining the agent's reachable space.

3.3.2 Pruning and Candidate Selection. Given the agent's local navigation tree at time step t , we select a set of candidate waypoints to guide future navigation.

First, predicted waypoints from previous time steps $1, \dots, t-1$ are transformed into the current local coordinate frame using the estimated agent pose and re-localized on the constructed navigation tree. Each re-localized waypoint is then associated with the corresponding tree sector it lies in.

Then, for leaf sectors that do not contain any re-localized waypoints, we apply a pruning strategy to filter invalid and structurally redundant candidates. Specifically, we first evaluate the validity of each sub-sector by computing its coverage ratio, defined as,

$$\rho(c'_i | c) = \frac{|c'_i|}{|c|}, \quad (12)$$

where $|\cdot|$ denotes the angular measure of a sector. A sub-sector c'_i is retained as a valid child node if its coverage ratio satisfies $\rho(c'_i | c) > k_1$, where $k_1 \in (0, 1)$ is a predefined threshold controlling the minimum proportion of navigable space inherited from the parent sector. To further reduce redundancy, we prune structurally overlapping waypoints based on the topology of the navigation tree. Let $\text{NCA}(w_i, w_j)$ denote the nearest common ancestor of waypoints w_i and w_j in the tree, and let $\text{depth}(\cdot)$ denote the depth of a node. For any pair of waypoints (w_i, w_j) , if

$$\max(\text{depth}(w_i), \text{depth}(w_j)) - \text{depth}(\text{NCA}(w_i, w_j)) \geq k_2, \quad (13)$$

where k_2 is a predefined threshold, then the two waypoints are considered structurally overlapping, and only their common ancestor is retained. This condition identifies sibling or near-sibling nodes that diverge only at shallow depth, indicating similar spatial regions. The proposed pruning strategy enforces a compact and non-redundant set of candidates by eliminating sibling waypoints that correspond to similar spatial regions, thereby preventing overlapping candidates.

Finally, for each remaining sector, we assign a candidate waypoint by selecting a representative node within the sector, ensuring sufficient coverage of newly discovered navigable directions.

3.3.3 Adaptive Step Length Refinement. Since the navigation tree is generated primarily from forward exploration information and lacks complete two-dimensional coverage, we further apply adaptive step length refinement after selecting the candidate waypoints. Specifically, we associate each candidate node v in the navigation tree with an exploration displacement vector,

$$\mathbf{s}(v) = [s^f, s^l, s^r], \quad (14)$$

which encodes the forward and lateral exploration distances reachable from v , where s^f denotes the forward extent and s^l, s^r denote the left and right lateral extents, respectively. For each candidate, we recursively compute the exploration displacement vectors of all its ancestor nodes $\pi(v)$ along the tree.

To adaptively determine the step length, we traverse the tree from the root and identify the deepest node v^* whose exploration displacement exhibits a significant change relative to its parent:

$$\|\mathbf{s}(v^*) - \mathbf{s}(\pi(v^*))\|_2 > k_3. \quad (15)$$

The depth of v^* is then selected as the refined step length for the current planning iteration, enabling longer steps in open regions while preserving finer granularity in cluttered environments.

3.4 Global Topo-Mapping and Agent Inference

After the aforementioned waypoint prediction, re-localized and predicted local waypoints are incrementally integrated into a global topological map, forming the discrete action space used for navigation decisions. Each node in the map is represented by a multimodal feature embedding that integrates visual, pose, and temporal information. The visual feature $\mathbf{f}^{\text{vis}}$ is initialized with the corresponding image embedding $\mathbf{v}_i$ and updated with a panoramic aggregated representation $\bar{\mathbf{v}}_i$ after node visitation. The pose feature is encoded using the agent's heading angle θ as $\mathbf{f}_i^{\text{pose}} = (\sin \theta_i, \cos \theta_i)$, and temporal information $\mathbf{f}_i^{\text{time}}$ encodes the corresponding time step for the visited nodes and is set to zero for unvisited nodes. Finally, each node is characterized by the following feature representation:

$$\mathbf{f}_i = (\mathbf{f}_i^{\text{vis}}, \mathbf{f}_i^{\text{pose}}, \mathbf{f}_i^{\text{time}}). \quad (16)$$

At each time step, the VLN agent selects a waypoint on this map as its navigation action, enabling long-horizon planning over a dynamically expanding topology.

We instantiate DIP² using ETP-Nav, a state-of-the-art VLN agent trained with cross-entropy supervision, as the navigation policy network. The agent takes as input the current topological map and the language instruction, and outputs a probability distribution over candidate waypoints. The selected waypoint is then executed by a

Table 1: Navigation results on R2R-CE in the Habitat simulator. NaVILA* corresponds to the original implementation trained exclusively on R2R-CE, to ensure fair comparison. SR serves as the primary metric. Best results are highlighted in bold.

Methods	Sim-to-Real	val_seen				val_unseen			
		NE ↓	OSR ↑	SR ↑	SPL ↑	NE ↓	OSR ↑	SR ↑	SPL ↑
Sim2Sim [14]	×	4.67	61	52	44	6.07	52	43	36
CWP-CMA [13]	×	5.20	61	51	45	6.20	52	41	36
CWP-RecBBRT [13]	×	5.02	59	50	44	5.74	53	44	39
GridMM [21]	×	4.21	69	59	51	5.11	61	49	41
BEVBert [1]	×	-	-	-	-	4.57	67	59	50
ETPNav [2]	×	3.95	72	66	59	4.71	65	57	49
CM ² [12]	✓	6.10	51	43	35	7.02	41.5	34.3	27.6
WS-MGMap [7]	✓	5.65	52	47	43	6.28	47.6	38.9	34.3
NaVid [24]	✓	-	-	-	-	5.47	49.1	37.4	35.9
ETPNav w/ FF [22]	✓	-	-	-	-	5.95	55.8	44.9	30.4
Uni-NaVid [23]	✓	-	-	-	-	5.58	53.3	47.0	42.7
NaVILA* [10]	✓	-	-	-	-	5.37	57.6	49.7	45.5
ETPNav w/ TB-WP (Ours)	✓	4.04	70.4	64.0	55.4	5.13	61.7	54.4	45.5

low-level controller to move the agent in the physical environment. Crucially, both training and inference are conducted using the same topological representation and action formulation in simulation and the real world. By enforcing invariance in perceptual inputs and decision spaces, DIP² eliminates domain-specific discrepancies and enables seamless sim-to-real transfer without modifying policy.

4 Experiments

In this section, we validate DIP² in both simulated and real-world environments. As discussed earlier, DIP² can be seamlessly integrated into existing agent architectures to facilitate sim-to-real transfer. Specifically, we adopt the classic ETP-Nav [2] as the base agent, denoted as ETP-Nav w/ DIP².

4.1 Experimental Setup

4.1.1 Simulated Environment. DIP² was evaluated in the Habitat Simulator using the R2R-CE [15] dataset. R2R-CE is constructed by converting the discrete trajectories of the R2R [4] dataset into continuous environments within the Habitat simulator [19].

4.1.2 Real-World Environment. We deployed DIP² on a real-world robotic platform. The robot was equipped with a surround-view camera system comprising four fisheye cameras, as well as a Livox 360 LiDAR sensor for localization and mapping. Following the simulated agent setting, high-level semantic decision-making was performed in a discrete manner, while low-level mapping and control were executed continuously throughout the navigation process. The real-world environment comprised three scenes: lounge, hallways, and office. For each scene, we collected 20, 15, and 15 navigation routes, respectively, and assigned a navigation instruction to each route, resulting in a total of 50 navigation episodes.

4.1.3 Baselines. The following state-of-the-art VLN-CE methods were selected for comparison. The first group consists of classic simulation-based agents, including Sim2Sim [14], CWP-CMA [13], CWP-RecBBRT [13], GridMM [21], BEVBert [1], ETPNav [2]. These methods rely on a panoramic RGB-D observation setting, which makes direct sim-to-real transfer challenging. Accordingly, we compared ETP-Nav w/ DIP² against these methods in simulated environments. The second group contains sim-to-real transferable agents, including CM² [12], WS-MGMap [7], NaVid [24], ETPNav w/ FF [22], Uni-NaVid [23], NaVILA [10]. These methods modify the conventional pipelines by adopting a monocular observation setting to facilitate sim-to-real transfer. Several of these methods were included in not only simulated but also real-world comparisons. We emphasize that DIP² is complementary to monocular methods, addressing a different axis of the sim-to-real gap focused on representation alignment rather than perceptual minimization.

4.1.4 Metrics. For the evaluation of navigation performance, the following metrics were adopted:

- **Navigation Error (NE)** is defined as the Euclidean distance between the agent’s final position and the destination.
- **Success Rate (SR)** denotes the proportion of successful navigation episodes, where success is defined as reaching the destination within a 3-meter threshold. This metric is adopted as the primary evaluation criterion.
- **Success weighted by Path Length (SPL)** adjusts the success rate by accounting for the efficiency of the trajectory, measured as the ratio between the ground-truth path length and the length of the path executed by the agent.
- **Oracle Success Rate (OSR)** reports the success rate under an oracle stopping policy.
- **Collision (Col.)** is defined as the proportion of tasks in which the agent collides with the environment.

Table 2: Navigation results in real-world environments. SR serves as the primary metric. Best results are highlighted in bold.

Methods	scene 1			scene 2			scene 3			Total.		
	Col. ↓	OSR ↑	SR ↑	Col. ↓	OSR ↑	SR ↑	Col. ↓	OSR ↑	SR ↑	Col. ↓	OSR ↑	SR ↑
CM ² [12]	25.0	25.0	25.0	26.7	26.7	13.3	20.0	26.7	20.0	24.0	26.0	20.0
WS-MGMap [7]	30.0	15.0	15.0	26.7	33.3	33.3	20.0	33.3	26.7	26.0	26.0	24.0
ETPNav w/ FF [22]	20.0	45.0	40.0	40.0	26.7	26.7	20.0	53.3	53.3	26.0	42.0	40.0
ETPNav w/ DIP ² (Ours)	20.0	55.0	50.0	26.7	46.7	33.3	13.3	73.3	66.7	20.0	58.0	50.0

4.2 Comparison with SOTA in Simulated Environments

The navigation performance of ETP-Nav with DIP² was first evaluated on the R2R-CE dataset in the Habitat simulator. As shown in Table 1, DIP² achieves substantially better performance than existing sim-to-real methods. Notably, DIP² improves the SR by at least 4.7 on the val_unseen split over prior sim-to-real approaches. These results highlight the strong capability of DIP² to understand natural language instructions and complete navigation tasks. This performance can be attributed to DIP²'s design choice to preserve the panoramic observation structure assumed during training, which is reconstructed from deployable sensors to acquire complete environmental semantics, thereby reducing mismatch in perception coverage between simulation and real-world deployment. Despite these gains, a performance gap remains between ETP-Nav w/ DIP² and its fully simulated counterpart, ETP-Nav. This discrepancy can be attributed to differences in the waypoint prediction pipeline. Specifically, ETP-Nav predicts waypoints directly from 3D point clouds and RGB semantic features, whereas DIP² projects all perceptual inputs onto a 2D occupancy map, retaining only structural information to facilitate sim-to-real transfer, and performs waypoint prediction solely based on this 2D representation. As a consequence, information loss in this projection is unavoidable, resulting in a minor performance drop with overall strong performance maintained.

4.3 Comparison with SOTA in Real-World Environments

Besides experiments in the Habitat simulator, our method was further evaluated in the real-world environment introduced in the experimental setup section. The results are shown in Table 2, where we report the Col., OSR, and SR for each of the three scenes as well as overall. These results demonstrate improved navigation success, consistent with the trends observed in simulated environments, thereby validating the effectiveness of DIP² in real-world deployment. Moreover, ETP-Nav w/ DIP² exhibits a lower collision rate, which can be attributed to DIP² explicitly modeling 2D occupancy information, thereby reducing the risk of collision. Furthermore, unlike simulated environments, where full 3D geometry is readily available and agents navigate in 3D space, most real-world robots operate on a mobile base and do not require full 3D information for navigable waypoint prediction. Consequently, performing waypoint prediction on a 2D occupancy map is particularly well suited to real-world settings, leading to safer and more reliable navigation.

Table 3: Ablation studies. SR serves as the primary metric. Best results are highlighted in bold.

Methods	StepLen.	val_unseen			
		NE ↓	OSR ↑	SR ↑	SPL ↑
DIP ² w/o SaMM	1.25	5.71	55.2	46.3	34.9
DIP ² w/o Prune	1.25	5.43	59.1	51.9	41.3
DIP ² w/o SLR	1.25	5.80	54.2	45.2	34.4
DIP ² w/o IWP	1.25	5.36	60.0	51.3	42.2
DIP ² _{StepLen.=1.00}	1.00	5.26	59.5	52.6	43.5
DIP ² _{StepLen.=1.25}	1.25	5.13	61.7	54.4	45.5
DIP ² _{StepLen.=1.50}	1.50	5.37	59.2	52.2	42.4
DIP ² _{StepLen.=2.00}	2.00	5.44	60.1	51.6	40.6

Additionally, we present qualitative results from real-world deployment. First, we illustrate examples of the synthesized pseudo-viewpoints in Figure 4, where we show several pairs of neighboring fisheye camera views along with their corresponding merged pseudo-viewpoints. The results demonstrate that the synthesized views are visually coherent and closely resemble real observations, indicating the effectiveness of the proposed view synthesis process. Furthermore, we visualize two representative navigation episodes in Figure 5, showcasing complete trajectories executed by ETP-Nav w/ DIP² in real-world environments. These examples illustrate that the agent is able to follow the given instructions accurately and navigate smoothly to the target locations, further confirming the robustness of our approach in real-world settings.

4.4 Ablation Studies

To provide a more comprehensive analysis and ensure reproducibility, we conducted ablation studies in simulated environments. Specifically, we evaluated the impact of the following components:

- DIP² w/o SaMM: removes the semantic-aided metric mapping module and instead directly projects point clouds within a predefined height threshold.
- DIP² w/o Prune: disables the pruning of structurally overlapping waypoints.
- DIP² w/o SLR: removes the step length refinement module.
- DIP² w/o IWP: without invariant waypoint prediction strategy (adopts ETPNav's waypoint predictor at training and our proposed strategy at inference).

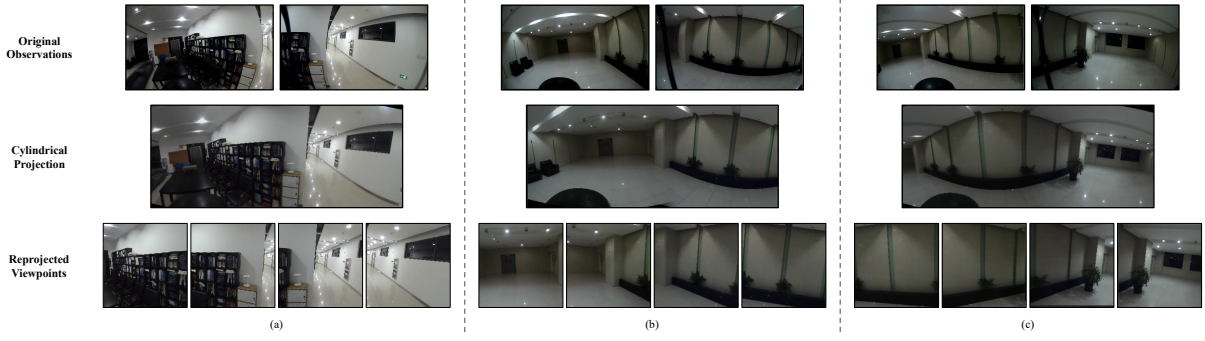


Figure 4: Qualitative visualization of the synthesized pseudo-viewpoints. For each example, we show two adjacent surround-view images, the projected partial panorama on the cylindrical surface, and the corresponding synthesized viewpoints.

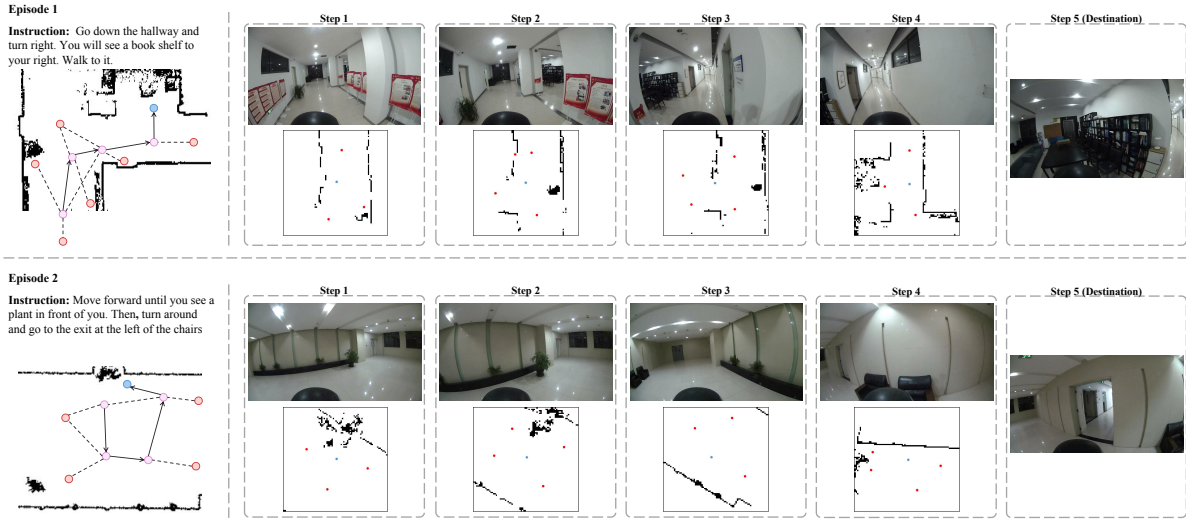


Figure 5: Qualitative results of two real-world navigation episodes using DIP². For each episode, we visualize the instruction and the overall trajectory. At each navigation step, we present the front-facing observation and the predicted local waypoints.

- $\text{DIP}^2_{\text{StepLen}}$: vary the maximum step length used for waypoint prediction.

As shown in Table 3, SaMM, Prune, SLR, and WIP each contribute positively to navigation performance, validating the effectiveness of the corresponding components. Regarding the choice of maximum step length, a step length of 1.25 achieves the best overall performance. Larger step lengths tend to result in insufficient spatial coverage, which reduces the success rate. In addition, they lead to significantly worse SPL, since longer step lengths produce coarser waypoint sequences that deviate more from the shortest-path trajectory, resulting in less efficient navigation. Conversely, smaller step lengths degrade SR as well, as they produce longer navigation episodes with more decision steps, increasing the likelihood of error accumulation and suboptimal actions.

5 Conclusion

In this paper, we propose DIP², a novel sim-to-real framework for VLN that emulates the idealized observation model with a realizable sensor setup while enforcing cross-domain representation

consistency. By unifying heterogeneous sensor inputs through invariant representations and removing bias in the decision pipeline, DIP² enables deployment under realistic sensing constraints without modification. This design decouples navigation reasoning from sensor-specific assumptions, allowing policies trained in simulation to transfer directly to real-world robots. Extensive results in both simulated and real environments demonstrate robust waypoint generation and reliable navigation. DIP² not only advances sim-to-real vision-and-language navigation by addressing observation-model mismatch and decision inconsistency, but is also broadly applicable to other embodied reasoning and navigation tasks that require invariant representations across heterogeneous sensing modalities.

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